

Slim Trabelsi

Mechanical Engineer

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[Portfolio](#)

Professional Summary

Results-driven mechanical engineer with **3 industrial internships**, **6+ engineering projects**, and extensive fabrication and technical leadership experience. Delivered a **15% optical efficiency gain** on a solar concentrator system and oversaw development of **2 competition-grade autonomous robots**; proficient in SolidWorks, CATIA, Abaqus, Ansys, and Python across design, simulation, and maintenance environments.

Work Experience

Facility Engineering Intern – Compressed Air Systems

Jun. 2026 – Aug. 2026(On going)

Marquardt

Tunisia

- Guided on-site diagnostics of a plant-wide compressed air network, identifying **3+ critical efficiency gaps** and documenting findings in structured technical reports.
- Compiled **2 corrective action plans** targeting energy fluctuations; standardized diagnostic templates and reduced field data collection time by **20%**.

Maintenance Engineering Intern

Jun. 2025 – Aug. 2025

Grande Fabrique De Confiserie Orientale (GFCO)

Tunisia

- Supervised **15+ preventive and corrective maintenance interventions** on **3 active production lines**, reducing unplanned stoppages by **18%** through structured root-cause analysis.
- Authored **2 maintenance workflow documents** adopted by the engineering team to streamline production scheduling and cut response times.

Part-Time Fabrication Technician and CAD Modeler

Jun. 2020 – Present

Future Pub – Metal Structures and Fabrication

Tunisia

- Delivered **10+ custom fabrication projects** alongside academic studies with **100% on-time delivery** and zero rework; generated SolidWorks CAD models reducing material waste by **15%**.
- Operated MAG welding, lathe, and milling equipment to manufacture components within tolerances of **plus or minus 0.5 mm**; managed on-site assembly for **10+ client sites**.

Volunteer Experience

Bureau Officer and Technical Lead – Robotics Club

2024 – 2026

National Engineering School of Tunis (ENIT)

Tunis, Tunisia

- Directed a national robotics competition with **100+ participants**, overseeing logistics, recruitment, and event execution; mentored **10+ junior engineers** across **5+ technical workshops**.
- Served as primary technical contact for **2 competition teams**, coordinating hardware integration and iterative testing under competition deadlines.

Projects

Final Year Project – Solar Thermal Concentrator System

2025 – 2026

National Engineering School of Tunis (ENIT)

Tunis, Tunisia

- Achieved **15% optical efficiency improvement** via compound parabolic concentrator geometry optimization, validated across **20+ ray-tracing simulation runs**; built full SolidWorks CAD model cutting prototyping time by **30%**.
- Conducted techno-economic analysis projecting a levelized cost of heat of **0.091–0.098 TND/kWh** and 14-year ROI; developed Python and JavaScript tools cutting manual analysis time by **40%**.

Autonomous Robotics – Competition Robots

2025 – 2026

ENIT Robotics Club

Tunis, Tunisia

- Engineered and programmed **2 autonomous robots** from concept to competition (ESP32 all-terrain and line-following); placed in **2 national competitions**.
- Integrated sensor arrays, motor drivers, and real-time C++ control logic achieving reliable autonomous navigation confirmed across **3+ test environments**.

Finite Element Analysis – Structural Component Study

Academic Project, ENIT

2024 – 2025

Tunis, Tunisia

- Modeled **3+ mechanical components** in SolidWorks and ran structural simulations in Abaqus and Ansys Mechanical to evaluate stress, strain, and deformation under service loads.
- Generated **3 technical reports** with verified failure analysis, material recommendations, and geometry modifications.

Education

Bachelor of Engineering – Mechanical Engineering

National Engineering School of Tunis (ENIT)

Sep. 2024 – Present

Tunis, Tunisia

- Coursework: Thermodynamics, Fluid Mechanics, Mechanical Design, FEA, Turbomachinery, Manufacturing Processes, Industrial Maintenance, Control Systems, Operations Research.

Pre-Engineering Degree – Physics and Technology

Preparatory Institute for Engineering Studies, El Manar

Sep. 2022 – Jun. 2024

Tunis, Tunisia

Awards

- **National Robotics Competition** – Competed with 2 self-built autonomous robots, 2025
- **National Event Organizer** – Directed ENIT Robots competition with 100+ participants, 2025–2026

Languages

- **Arabic** – Native **English** – TOEIC (B2) **French** – DELF (B2)

Skills

CAD and Simulation: SolidWorks, CATIA, Abaqus, Ansys Mechanical, MATLAB

Engineering: Mechanical Design, Finite Element Analysis, Structural Analysis, Fracture Mechanics, Thermodynamics, Fluid Mechanics, Turbomachinery, Preventive and Corrective Maintenance, Root-Cause Analysis

Programming and Tools: Python, C, C++, Arduino, ESP32, JavaScript, HTML, CSS

Soft Skills: Analytical thinking, problem-solving, teamwork, leadership, project management, technical communication, Technical reporting, cross-functional coordination, event planning, deadline management